

Alessandro Farace di Villaforesta

ABOUT

Full-stack engineer and solo founder of Janus, an AI flashcard generator with **3.5k registered users**, **150+ paying customers**, and **250k+ exported cards**. **Cambridge** Computer Science graduate with **1st class honours** and two years of production **low-latency C++** experience. Spent the past three months developing an extensible, open source agent runtime that can be embedded into any Typescript app.

SKILLS

- **Languages:** TypeScript, Python, C++
- **Systems:** Boost.Asio, gRPC, CMake, Tracy Profiler
- **Product Engineering:** React, TanStack Query, FastAPI, Tailwind CSS, Electron, Webpack
- **Machine Learning:** PyTorch, PyTorch Geometric, scikit-learn, NumPy, pandas

WORK

Founder, Janus

2024-2026

- Solo-bootstrapped an AI flashcard generator for medical students, turning PDFs and notes into structured flashcards for spaced repetition apps like Anki.
- Scaled to **3.5k registered users** with **150+ paying customers** and **250k+ exported cards**.
- Built the product end-to-end across TypeScript, React, TanStack Query, Python, and FastAPI, including document ingestion, markdown highlighting, optimistic updates, export workflows, and a model-agnostic generation pipeline.

C++ Developer, Pulsar Trading

2022-2024

- Built C++ infrastructure for low-latency market-data feeds across **20+ exchanges**.
- Designed and implemented a connection scheduling framework to detect & auto-heal stale connections and minimize throttling.
- Led an initiative that reduced the “line latency” by up to **100 microseconds in the median** and several hundred microseconds in the mean.

Supervisor, University of Cambridge

October-December 2023

- Remotely tutored 10 second-year Cambridge students in the C & C++ course.

Intern, Arqit

Summer 2022

- Worked in the Innovation Team, exploring possible applications of their post-quantum cryptography in blockchains.
- Used the Tamarin Prover to verify security protocols.

EDUCATION

University of Cambridge, 1st Class, Computer Science

2019-2022

- **King's Scholar** for receiving a **1st** in all exams. Ranked 11th out of 123 in second year.
- Part of the inaugural cohort of the **King's Entrepreneurship Lab**.

St Paul's School

2013-2019

- 4 A* at A-level and 11 A* at GCSE.
- Senior scholar and prize winner in Computer Science and Mathematics.

PROJECTS

Franklin

April 2026-Present

- Building an open-source TypeScript agent runtime with Obsidian as the first host application.
- Designed an extension architecture for app-owned tools, environment/sandbox interaction, and shared UI state.
- Recruited a university friend to collaborate full-time with me.

Explainable AI for Cancer Diagnosis

September 2021-May 2022

- Developed an automated cancer diagnosis program that makes predictions on H&E histological images.
- Implemented HoVerNet to generate cell graphs, leveraged a Graph Convolutional Network to predict the presence of cancer and GCExplainer to justify predictions.
- Worked under the supervision of Professor Pietro Liò, Lucie Charlotte Magister, and Pietro Barbiero.